

Calculus I Lesson 43B

Definition of e : e is a real number such that $\ln e = \int_1^e \frac{1}{x} \cdot dx = 1$.

Note: a) $\ln e^x = x \cdot \ln e = x \cdot 1 = x$.

b) If $y = e^x$, then $\ln y = \ln e^x = x$

c) $D_x(\ln y) = D_x(x)$

Using implicit differentiation: $\frac{1}{y} \cdot D_x(y) = 1 \Rightarrow D_x(y) = y \Rightarrow D_x(e^x) = e^x$

d) $\int e^x \cdot dx = e^x + C$

1) Find $D_x(e^{2x-3})$.

a) Let $u = 2x - 3$. $\frac{du}{dx} = \underline{\hspace{2cm}}?$

b) Using Chain Rule: $D_x(e^{2x-3}) = D_x(e^u) = e^u \cdot \frac{du}{dx} = \underline{\hspace{2cm}}?$

2) Find $D_x(e^{4x})$.

a) Let $u = 4x$. $\frac{du}{dx} = \underline{\hspace{1cm}}?$

b) Using Chain Rule: $D_x(e^{4x}) = D_x(e^u) = e^u \cdot \frac{du}{dx} = \underline{\hspace{1cm}}?$

3) Find $D_x(e^{x^2-4})$.

a) Let $u = x^2 - 4$. $\frac{du}{dx} = \underline{\hspace{2cm}}?$

b) Using Chain Rule: $D_x(e^{x^2-4}) = D_x(e^u) = e^u \cdot \frac{du}{dx} = \underline{\hspace{2cm}}?$

4) Find $D_x(3e^{\sqrt{x}})$.

a) Let $u = \sqrt{x}$. $\frac{du}{dx} = \underline{\hspace{1cm}}?$

b) Using Chain Rule: $D_x(3e^{\sqrt{x}}) = D_x(3 \cdot e^u) = 3 \cdot D_x(e^u) = 3 \cdot e^u \cdot \frac{du}{dx} = \underline{\hspace{1cm}}?$

5) Find $D_x(e^x + e^{-x})$.

a) $D_x(e^{-x}) = \underline{\hspace{1cm}}?$ Hint: Let $u = e^{-x}$ and use the Chain Rule

b) $D_x(e^x + e^{-x}) = \underline{\hspace{1cm}}?$

6) Evaluate $\int e^{4x} \cdot dx$.

a) Let $u = 4x$. $\frac{du}{dx} = \underline{\hspace{1cm} ? \hspace{1cm}}$ and $\frac{1}{4} \cdot du = \underline{\hspace{1cm} ? \hspace{1cm}}$

b) Write the indefinite integral $\int e^{4x} \cdot dx$ in terms of u and then evaluate.

$$\int e^{4x} \cdot dx = \int (\underline{\hspace{1cm} ? \hspace{1cm}}) \cdot du = \underline{\hspace{1cm} ? \hspace{1cm}}$$

c) $\int e^{4x} \cdot dx = \underline{\hspace{1cm} ? \hspace{1cm}}$ (Note: Answer must be in terms of x .)

7) Evaluate $\int e^{x^2} \cdot x \cdot dx$.

a) Let $u = x^2$. $\frac{du}{dx} = \underline{\hspace{1cm}}?$ and $\frac{1}{2} \cdot du = \underline{\hspace{1cm}}?$

b) Write the indefinite integral $\int e^{x^2} \cdot x \cdot dx$ in terms of u and then evaluate.

$$\int e^{x^2} \cdot x \cdot dx = \int (\underline{\hspace{1cm}}?) \cdot du = \underline{\hspace{1cm}}?$$

c) $\int e^{x^2} \cdot x \cdot dx = \underline{\hspace{1cm}}?$ (Note: Answer must be in terms of x .)

8) Evaluate $\int \frac{e^{1/x}}{x^2} \cdot dx$.

Hint: $\int \frac{e^{1/x}}{x^2} \cdot dx = \int e^{1/x} \cdot x^{-2} \cdot dx$

a) Let $u = \frac{1}{x} = x^{-1}$. $\frac{du}{dx} = \underline{\hspace{1cm}}?$ and $-1 \cdot du = \underline{\hspace{1cm}}?$

b) Write the indefinite integral $\int \frac{e^{1/x}}{x^2} \cdot dx$ in terms of u and then evaluate.

$$\int \frac{e^{1/x}}{x^2} \cdot dx = \int e^{1/x} \cdot x^{-2} \cdot dx = \int (\underline{\hspace{1cm}}?) \cdot du = \underline{\hspace{1cm}}?$$

c) $\int \frac{e^{1/x}}{x^2} \cdot dx = \underline{\hspace{1cm}}?$ (Note: Answer must be in terms of x .)

9) Evaluate $\int \frac{e^x}{e^x + 4} \cdot dx$.

Hint: $\int \frac{e^x}{e^x + 4} \cdot dx = \int \frac{1}{e^x + 4} \cdot e^x \cdot dx$

a) Let $u = e^x + 4$. $\frac{du}{dx} = \underline{\hspace{1cm}} ? \hspace{1cm}$ and $du = \underline{\hspace{1cm}} ? \hspace{1cm}$

b) Write the indefinite integral $\int \frac{e^x}{e^x + 4} \cdot dx$ in terms of u and then evaluate.

$$\int \frac{e^x}{e^x + 4} \cdot dx = \int \frac{1}{e^x + 4} \cdot e^x \cdot dx = \int (\underline{\hspace{1cm}} ? \hspace{1cm}) \cdot du = \underline{\hspace{1cm}} ? \hspace{1cm}$$

c) $\int \frac{e^x}{e^x + 4} \cdot dx = \underline{\hspace{1cm}} ? \hspace{1cm}$ (Note: Answer must be in terms of x .)

10) Evaluate $\int \frac{e^x - e^{-x}}{e^x + e^{-x}} \cdot dx$

Hint: $\int \frac{e^x - e^{-x}}{e^x + e^{-x}} \cdot dx = \int \frac{1}{e^x + e^{-x}} \cdot (e^x - e^{-x}) \cdot dx$

a) Let $u = e^x + e^{-x}$. $\frac{du}{dx} = \underline{\quad ? \quad}$ and $du = \underline{\quad ? \quad}$

b) Write the indefinite integral $\int \frac{e^x - e^{-x}}{e^x + e^{-x}} \cdot dx = \int \frac{1}{e^x + e^{-x}} \cdot (e^x - e^{-x}) \cdot dx$
in terms of u and then evaluate.

$$\int \frac{1}{e^x + e^{-x}} \cdot (e^x - e^{-x}) \cdot dx = \int (\underline{\quad ? \quad}) \cdot du = \underline{\quad ? \quad}$$

c) $\int \frac{e^x - e^{-x}}{e^x + e^{-x}} \cdot dx = \underline{\quad ? \quad}$ (Note: Answer must be in terms of x .)

